

Question:
John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for \$15 each. He ends up buying 8 video games for \$20 each and has \$5 left. How many lego sets does he still have?

Reasoning + Answer:

Let's carefully analyze the problem step by step .

John has **13** Lego sets and sells them all for **\$15** each .

So , total money from selling Legos :

$$13 \times 15 = 195 \text{ dollars}$$

He uses this money to buy **8** video games at **\$20** each , and has **\$5** left .

So , money spent on video games :

$$8 \times 20 = 160 \text{ dollars}$$

Money left after purchase :

$$5 \text{ dollars}$$

Total money available :

$$160 + 5 = 165 \text{ dollars}$$

But wait he sold all his Legos , so he should have received \$195 . However , the problem says he ends up with \$5 left after buying 8 video games .

So , total money from Legos = money spent on games + money left :

$$\text{Total money from Legos} = 160 + 5 = 165$$

But he sold **13** sets at **\$15** each , which gives :

$$13 \times 15 = 195$$

So , there's a contradiction unless the number of Legos sold is less than 13 .

Wait the question is : **How many Lego sets does he still have ?**

This implies that he may not have sold all of them .

But the first sentence says : "John plans to sell all his toys and use the money to buy video games ." Then says he sells them but does he sell all ?

But then it says he ends up buying 8 video games and has \$5 left .

So , total money from selling legos = 15 × number of sets sold .

Let's suppose he sold **x** Lego sets .

Then total money from sales = 15x

Money spent on video games = 8 × 20 = 160

Money left = \$5

$$15x = 160 + 5 = 165$$

$$x = 165 / 15 = 11$$

So , he sold **11** Lego sets .

He originally had **13** sets .

Therefore , the number of Lego sets he still has is :

$$13 - 11 = 2$$

Final Answer :

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